

MATH120-24S2 Discrete Mathematics

Tutorial 6

References

- Topic 7 and 8: Modular arithmetic
- Section 8.4 of Epp

Work on the following problems during the tutorial in week 7.

1. [M] Determine the total number of suitable affine ciphers (i.e. these that are a permutation) for an alphabet of

⇒

- (a) 24 letters,
- (b) p letters where p is a prime,
- (c) 33 letters,

Solution:

Recall from lectures that with the Affine cipher $e(x) = ax+b$, the number of possible ciphers for a given alphabet is the number of possible choices for a multiplied by the number of possible choices for b .

- (a) In an alphabet of 24 letters there are 24 choices for b . We need to determine the number of invertible elements in $\mathbb{Z}_{24}$: these are precisely the elements between 1 and 23 that are coprime to 24: 1, 5, 7, 11, 13, 17, 19, 23. Hence, there are 8 invertible elements in $\mathbb{Z}_{24}$ and thus $24 \times 8 = 192$ possible Affine ciphers in an alphabet of 24 letters. *Note that the useless encryption $e(x) = x$ is one of these possible affine ciphers.*
- (b) In an alphabet of p letters there are p choices for b . Recall that a must be invertible. We have seen that every non-zero element in $\mathbb{Z}_p$ is invertible, so there are $p - 1$ choices for a . Hence there are $p(p - 1) = p^2 - p$ possible Affine ciphers in an alphabet of p letters. *Note that the useless encryption $e(x) = x$ is one of these possible affine ciphers.*

- (c) In an alphabet of 33 letters there are 33 choices for b . Recall that a must be invertible. We have seen that if $n = p \cdot q$ for two distinct primes, then $\phi(n) = (p - 1)(q - 1)$. The Euler ϕ function $\phi(n)$ counts precisely the number of invertible elements in $\mathbb{Z}_n$. Since $p = 3$ and $q = 11$ are prime, $\phi(33) = 20$.

Hence there are $33 \times 20 = 660$ possible Affine ciphers in an alphabet of 33 letters. *Note that the useless encryption $e(x) = x$ is one of these possible affine ciphers.*

$\Rightarrow$ 2. [B-M] The ciphertext

R W J W Q T O O M Y H K U X G O E M Y P

was encrypted with an affine cipher in $\mathbb{Z}_{26}$, using the usual encoding of the alphabet. Given that the plaintext letters E, T are encrypted as the ciphertext letters W, X respectively, determine

- (a) the encryption function e ,
- (b) the decryption function d , and
- (c) the original plaintext message.

Solution:

- (a) We are given that the plaintext letter E (04) equates to W (22) and T (19) equates to X (23). The Affine cipher, or encryption function, is of the form $e(x) = ax + b$, therefore, we have

$$4a + b = 22 \tag{1}$$

$$19a + b = 23. \tag{2}$$

Subtracting (1) from (2), we get

$$15a = 1$$

in $\mathbb{Z}_{26}$. Hence $a = 15^{-1}$ in $\mathbb{Z}_{26}$. From the table in Example 9.5 in the lecture notes, we know that $a = 15^{-1} = 7$ (you could also have worked this out using the method given by the proof of Theorem 8.5

in the lecture notes, or using trial and error). Substituting $a = 7$ into 1, we find that

$$\begin{aligned} b &= 22 - 4(7) \\ &= 22 - 28 \\ &= -6 \end{aligned}$$

in $\mathbb{Z}_{26}$. Consequently, the encryption function is

$$e(x) = 7x + 20.$$

- (b) The decryption function has the form $d(y) = a^{-1}(y - b)$. From part (i) we have that $b = 20$ and that $a = 15^{-1} = 7$ and hence $a^{-1} = 7^{-1} = 15$. It follows that the decryption function is

$$d(y) = 15(y - 20).$$

- (c) Noting that

$$\begin{aligned} R &= 17, & W &= 22, & J &= 09, & Q &= 16, & T &= 19, \\ O &= 14, & M &= 12, \\ Y &= 24, & H &= 07, & K &= 10, & U &= 20, & X &= 23, \\ G &= 06, & E &= 04, & P &= 15 \end{aligned}$$

and then applying $d(y)$ to each letter above, we have

$d(17)=15(17-20)=7$, $d(22)=15(22-20)=4$, $d(09)=15(09-20)=17$,
 $d(16)=15(16-20)=18$, $d(19)=15(19-20)=11$, $d(14)=15(14-20)=14$
 $d(12)=15(12-20)=10$, $d(24)=15(24-20)=8$, $d(07)=15(7-20)=13$,
 $d(10)=15(10-20)=6$, $d(20)=15(20-20)=0$, $d(23)=15(23-20)=19$,
 $d(06)=15(6-20)=24$, $d(04)=15(4-20)=20$, $d(15)=15(15-20)=3$
 in $\mathbb{Z}_{26}$. Converting these numbers back to letters of the alphabet, we recover the original plaintext message:

H E R E S L O O K I N G A T Y O U K I D.

3. [B-M] You have intercepted the following message in ciphertext:

Y F X M P C E S P Z C J T D F D P Q F W Q
Z C P Y N T A S P C T Y R X P D D L R P D

You know that it has been enciphered using a Caesar cipher, but you don't know the key. Your task is to recover the original message. Here are the steps you must carry out:

- Count the number of occurrences of each letter.
- Based on the results of (a), use frequency analysis to recover the key.
- Decrypt the message using the Caesar cipher and the key you obtained in (b).

Solution:

- Counting the number of occurrences of each letter gives:

A	B	C	D	E	F	G	H	I	J	K	L	M	N	O	P	Q	R	S	T	U	V	W	X	Y	Z
1	0	4	5	1	3	0	0	0	1	0	1	1	1	0	7	2	2	2	3	0	0	1	2	3	2

- The most frequently occurring letter is P, so our initial guess is that E encrypts to P (since E is the most frequently occurring letter in written English). If this is the case then the key is the difference between the letters P (15) and E (04) in $\mathbb{Z}_{26}$, which is 11. We can see this by simply counting, or noting that

$$e(04) = 04 + b = 15$$

and solving for b .

- If $b = 11$, then the Caesar cipher encrypts letters of the alphabet as follows:

A	B	C	D	E	F	G	H	I	J	K	L	M	N	O	P	Q	R	S	T	U	V	W	X	Y	Z
P	Q	R	S	T	U	V	W	X	Y	Z	A	B	C	D	E	F	G	H	I	J	K	L	M	N	O

Hence the messages deciphers to

NUMBER THEORY IS USEFUL FOR ENCIPHERING MESSAGES.

- $\Rightarrow$ 4. [M] Use Euler's theorem to determine the remainder of 2^{2023} when divided by 35.

Solution:

Since $35 = 5 \cdot 7$ and $5, 7$ are prime, we have $\phi(35) = 4 \cdot 6 = 24$. Euler's theorem tells us that $2^{24} \equiv 1 \pmod{35}$.

So in order to calculate $2^{2023} \pmod{35}$, we can write

$$2^{2023} = 2^{84 \cdot 24 + 7} \equiv 2^7 \equiv 128 \equiv 23 \pmod{35}$$

5. [M] You are given the two primes $p = 3$ and $q = 11$ in order to construct your own RSA cryptosystem.

- (a) What are the values of n and $\phi(n)$?
- (b) You choose your public key $e = 5$. However, this turns out to be a bad choice. Explain why.
- (c) You try again, and this time choose your public key $e = 7$. What is your private key d ?
- (d) Your good friend Bob uses your public key to encrypt the following message (already in numeric form):

16

and sends it to you. What is the ciphertext you receive?

- (e) Sneaky Eve intercepts Bob's message and manages to decipher it. Explain how she did it. (You may assume that you have successfully kept p, q and d a secret and that Eve knows only n and e .)

Solution:

- (a) $n = 3 \cdot 11 = 33$ and $\phi(n) = 2 \cdot 10 = 20$.
- (b) $e = 5$ is not invertible in $\mathbb{Z}_{20}$ since $\gcd(5, 20) = 5 \neq 1$.
- (c) We need to find the inverse of 7 in $\mathbb{Z}_{20}$. We can use the extended Euclidean algorithm for this: $20 = 2 \cdot 7 + 6$, and $7 = 1 \cdot 6 + 1$, so $1 = 7 - 6 = 7 - (20 - 2 \cdot 7) = 3 \cdot 7 - 20$. We see that the inverse of 7 in $\mathbb{Z}_{20}$ is 3.
- (d) Bob sends $16^7 \pmod{33}$, which is $2^{28} = 2^{20} \cdot 2^8 \equiv 1 \cdot 2^8 = 256 = 7 \cdot 33 + 25 \equiv 25 \pmod{33}$. We have used Euler's theorem to write $2^{20} \equiv 1 \pmod{33}$.

(e) Since $n = 33$ is very small, Eve can easily find the factors $p = 3$ and $q = 11$ of n . Since Eve knows p and q , she knows $\phi(n) = (p - 1)(q - 1) = 20$ and she can calculate the private key d by finding the inverse of the public key 7 modulo 20. Once she has the private key $d = 3$, she simply calculates $25^3 \pmod{35} \equiv 16$ and retrieves the message.

6. [C] Alice sends a message to Bob encrypted using RSA encryption. Suppose Eve somehow finds out what the value of $\phi(n)$ is. Show that Eve can now retrieve Alice's message.

Solution:

Eve now knows n , e , and $\phi(n)$. Using the Euclidean algorithm, she can calculate $d = e^{-1} \pmod{\phi(n)}$, and hence can decrypt just like Bob can.

Alternatively,

Suppose that Eve knows n and $\phi(n)$. Since $n = pq$ and $\phi(n) = (p - 1)(q - 1)$, we have that $\phi(n) = pq - (p + q) + 1$.

So she needs to find p and q with

$$pq = n \quad \text{and} \quad p + q = n - \phi(n) + 1.$$

This shows that p and q are the roots of the quadratic equation

$$X^2 - (n - \phi(n) + 1)X + n.$$

(Recall that the roots X_1, X_2 of the equation $X^2 + bX + c$ satisfy $X_1 + X_2 = -b$ and $X_1X_2 = c$. If you haven't seen this before, make sure you understand why this is true!)

So since n and $\phi(n)$ are known to Eve, she can simply plug in these values and solve the quadratic equation (using the quadratic formula)

$$X^2 - (n - \phi(n) + 1)X + n$$

to retrieve p and q , and rebuild the whole system from them.

7. [M-C] One can show that

$$\phi(n) = n \prod_{p|n} \left(1 - \frac{1}{p}\right), \quad (3)$$

where p is prime (i.e. the product ranges over all possible prime divisors of n .)

- (a) Use (3) to find $\phi(p)$, p prime, and $\phi(pq)$, p, q prime, $p \neq q$.
- (b) Use (3) to show that $\phi(p^k) = p^k - p^{k-1}$, when p is prime and k is a positive integer.
- (c) Use (3) to show that if $\gcd(m, n) = 1$, $\phi(mn) = \phi(m)\phi(n)$.
- (d) Now assume that $\psi(n)$ is a function that satisfies
 - i. $\psi(p^k) = p^k - p^{k-1}$ for a prime p and a positive integer k ,
 - ii. $\psi(mn) = \psi(m)\psi(n)$ if $\gcd(m, n) = 1$.

Show that

$$\psi(n) = n \prod_{p|n} \left(1 - \frac{1}{p}\right),$$

where p is prime.

Solution:

- (a) If n is a prime, there is only one prime, namely n dividing n . So we find: $\phi(n) = n(1 - \frac{1}{n}) = n - 1$ which was what we needed to show. If $n = pq$, where p and q are distinct primes, there are precisely two primes, p and q dividing n . So we find: $\phi(n) = n(1 - \frac{1}{p})(1 - \frac{1}{q}) = \frac{n}{pq}(p-1)(q-1) = (p-1)(q-1)$ which was what we needed to show.
- (b) The only prime divisor of p^k is p , so we find $\phi(p^k) = p^k(1 - \frac{1}{p}) = p^k - p^{k-1}$.
- (c) If $\gcd(m, n) = 1$, then m and n have no mutual prime divisors. Let $m = p_1^{e_1} \dots p_s^{e_s}$ and $n = q_1^{f_1} \dots q_t^{f_t}$ where p_i, q_j are *mutually distinct* primes.

$$\begin{aligned} \phi(mn) &= mn \prod_{p|mn} \left(1 - \frac{1}{p}\right) = mn \prod_{p \in \{p_1, \dots, p_s, q_1, \dots, q_t\}} \left(1 - \frac{1}{p}\right) \\ &= m \prod_{p \in \{p_1, \dots, p_s\}} \left(1 - \frac{1}{p}\right) n \prod_{p \in \{q_1, \dots, q_t\}} \left(1 - \frac{1}{p}\right) = \phi(m)\phi(n). \end{aligned}$$

- (d) Note that it follows from $\psi(mn) = \psi(m)\psi(n)$ if $\gcd(m, n) = 1$ that $\psi(n_1 n_2 \dots n_s) = \psi(n_1)\psi(n_2) \dots \psi(n_s)$ when $\gcd(n_i, n_j) = 1$ for all $i \neq j$. Now let $n = p_1^{e_1} \dots p_s^{e_s}$, where p_i are mutually distinct primes. This means that $\gcd(p_i, p_j) = 1$ if $i \neq j$. Then $\psi(n) = \psi(p_1^{e_1} \dots p_s^{e_s}) = \psi(p_1^{e_1})\psi(p_2^{e_2}) \dots \psi(p_s^{e_s})$, which in turn equals $(p_1^{e_1} - p_1^{e_1-1})(p_2^{e_2} - p_2^{e_2-1}) \dots (p_s^{e_s} - p_s^{e_s-1})$ by (i). This equals
- $$p_1^{e_1} \dots p_s^{e_s} \left(1 - \frac{1}{p_1}\right) \left(1 - \frac{1}{p_2}\right) \dots \left(1 - \frac{1}{p_s}\right) = n \prod_{p|n} \left(1 - \frac{1}{p}\right).$$